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Review article

A review on online state of charge and state of health estimation for lithium-ion batteries in electric vehicles

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ABSTRACT

With electric vehicles (EVs) being widely accepted as a clean technology to solve carbon emissions in modern transportation, lithium-ion batteries (LIBs) have emerged as the dominant energy storage medium in EVs due to their superior properties, like high energy density, long lifespan, and low self-discharge. Performing real-time condition monitoring of LIBs, especially accurately estimating the state of charge (SOC) and state of health (SOH), is crucial to keep the LIBs work under safe state and maximize their performance. However, due to the non-linear dynamics caused by the electrochemical characteristics in LIBs, the accurate estimations of SOC and SOH are still challenging and many technologies have been developed to solve this challenge. This paper reviews and discusses the state-of-the-art online SOC and SOH evaluation technologies published within the recent five years in view of their advantages and limitations. As SOC and SOH are strongly correlated, the joint estimation methods are specifically reviewed and discussed. Based on the investigation, this study eventually summarizes the key issues and suggests future work in the real-time battery management technology. It is believed that this review will provide valuable support for future academic research and commercial applications.

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List of nomenclature

EVs	Electric vehicles
LIBs	Lithium-ion batteries
BMS	Battery management system
SOC	State of charge
SOH	State of health
SOE	State of energy
SOP	State of power
SOT	State of temperature
SOS	State of safety
KF	Kalman filter
OCV	Open circuit voltage
AHC	Ampere-hour counting
DA	Differential analysis
SEI	Solid electrolyte interphase
EOL	End of life
EM	Empirical model
ECM	Electrochemical model
EECM	Electrical equivalent circuit model
ECIM	Electrochemical impedance model
PNGV	Partnership for a new generation of vehicles
FOM	Fractional-order model
2RCH	2RC with one-state hysteresis
ETNN	Electrochemical–thermal–neural-network
HPPC	Hybrid pulse power characterization
CCD	Constant current discharge
FUDS	Federal urban driving schedule
DST	Dynamic stress test
UDDS	Urban dynamometer driving schedule
BJDC	Beijing driving cycle
PIMs	Parameter identification methods
GA	Genetic algorithm
RLS	Recursive least squares
FFRLS	Forgetting factor recursive least squares
PSO	Particle swarm optimization
RRTLS	Recursive restricted total least squares
PF	Particle filter
SMO	Slide mode observer
EKF	Extended Kalman filter
AEKF	Adaptive extended Kalman filter
IAEKF	Intelligent adaptive extended Kalman filter
CDKF	Central difference Kalman filter
SRCDKF	CDKF with square root second-order difference transform
UKF	Unscented Kalman filter
IUKF	Improved unscented Kalman filter
AUKF	Adaptive unscented Kalman filter
CKF	Cubature Kalman filter
ACKF	Adaptive cubature Kalman filter
AFDCKF	Adaptive fifth-degree cubature Kalman filter
ASRCKF	Adaptive square root cubature Kalman filter
UPF	Unscented particle filter
IAPF	Improved adaptive particle filter
HF–UKF	H infinity and UKF
DKF	Dual Kalman filter
DEKF	Dual extended Kalman filter
CPF	Cubature particle filter
ANN	Artificial neural network
SVM	Support vector machine
SVR	Support vector regression
GPR	Gaussian process regression
CGA	Chaos genetic algorithm
IGGA	Improved Chaos genetic algorithm
DNN	Deep feedforward neural networks
RNN–CNN	Recursive neural network and convolutional neural network
GRU–RNN	Recurrent neural network with gated recurrent unit
DBN	Deep belief network
FFNN	Feedforward neural network
GRU–GPR	GPR with gated recurrent unit kernel
LSTM	Long short-term memory
B-LSTM	Bidirectional long short-term memory
SBLSTM	Stacked bidirectional long short-term memory
ICA	Incremental capacity analysis
DVA	Differential voltage analysis
DTV	Differential thermal voltammetry
DMP	Differential mechanical parameter
SPM	Single particle model
SPMe	Single particle model with electrolyte
eSPM	Enhanced single particle model
P2D-SPM	Pseudo-two-dimensional model coupled with single particle model
ELM	Extreme learning machine
MELM	Metabolic extreme learning machine
PSO-LSSVR	Particle swarm optimization-least square support vector regression
CC–CV	Constant current–constant voltage
LSSVM	Least squares support vector machine
GRU–CNN	Gate recurrent unit–convolutional neural network
VLR-LSTM	Variable-length-input long short-term memory
AST-LSTM	A variant of LSTM
RMSE	Root-mean-squared-error
MAE	Mean absolute error
MAX	Maximum relative error
MSE	Maximum relative error and mean square error
SVSF	Smooth variable structure filter
BS-SRCKF	Backward smoothing square root cubature Kalman filter
EIS	Electrochemical impedance spectroscopy

1. Introduction

Dealing with the pressure from environmental damage and energy crisis has been one important task for all countries (Akinlabi and Solyali, 2020). Electric vehicles (EVs) have been widely accepted as a clean transportation technology to reduce the reliance on fossil fuels, and play an important role in slowing down global warming rate thanks to the exploitation of the sustainable energy (Wang et al., 2020d; Al-Ghussain et al., 2021), and the development of energy management technologies (Gong et al., 2020; Lan et al., 2021). As the energy power for the EVs, batteries are the most critical part in the performance and safe running of EVs. A variety of rechargeable batteries are developed as energy storage for EVs in which lithium-ion batteries (LIBs) become the dominant power storage solution, owing to their unique merits such as high density and long lifespan (Guo et al., 2021).

Developing advanced battery management system (BMS) for EVs has been a popular research topic due to its importance and existing challenges. On the one hand, the high penetration of EVs brings significant impact and challenges to the power grid (Min et al., 2021). Currently, the hybrid AC/DC microgrids combined with renewable energy sources such as solar energy and wind power have been developed as power sources for EVs (Wang et al., 2020b). A reliable BMS can provide accurate state estimations and ensure the safety of batteries, which can facilitate the optimal operation and management of the distributed grids. On the other hand, many uncertainties in the practical operations of EVs bring serious challenges to the BMS (Mohamed et al., 2021). For example, the battery systems not only serve to drive the electric motor, but also supply power to other electronic systems. EVs often work in complex working conditions, such as frequent acceleration and deceleration and the charging behavior from humans is often random. Furthermore, the battery is an electrochemical system so that the high nonlinearity and time-varying characteristics make the state estimation very challenging (Ee et al., 2021). Therefore, developing accurate and reliable technologies in BMS is still a demanding task to ensure batteries and the related energy systems work in a safe state and maximize their performance.

Battery management technologies involve various types of estimations, such as the state of charge (SOC), state of health (SOH), state of energy (SOE), state of power (SOP), state of temperature (SOT), and state of safety (SOS). Generally, the strongly correlated SOC and SOH monitoring are the main concerns and the basis to improve reliability and ensure safety (Hu et al., 2018b). The SOC estimation of LIBs aims to check the remaining capacity of a battery during a charge–discharge cycle, which can avoid the overcharging and overdischarging of the battery. In particular, the battery SOC changes with time when charging/discharging and it is an important factor for further SOH prediction. The SOH estimation is to predict the remaining useful life or the remaining charge–discharge cycles, which infers if LIBs need to be replaced with new ones (Zhang et al., 2018). In contrast, the battery SOH characterized by the slow-changing parameters, such as capacity fading and resistance increasing, varies with cycles and hence it needs to be monitored in a long timescale (Hu et al., 2019). However, both SOC and SOH cannot be directly measured by the sensors, they are only monitored and reflected based on the measured parameters such as voltage, current, temperature and internal resistance (Shrivastava et al., 2019). Due to the electrochemical dynamics inside the battery, the accurate SOC/SOH estimation of the battery remains as a challenging work.

To date, a few review papers on SOC and SOH estimation of LIBs have been published. Hannan et al. (2016) reviewed the SOC estimation of LIBs based on battery model, estimation methods and their merits and drawbacks (Hannan et al., 2017). Zheng et al. (2018b) focused on the error sources affecting SOC estimation but the review on estimation method was not comprehensive. Prashant et al. (2019) just elaborated the overview of online SOC estimation based on Kalman filter (KF) family algorithm and presented the key challenges (Shrivastava et al., 2019). Lipu et al. (2020) discussed the data-driven SOC estimation methods of LIBs in EV applications and did not focus on the model-based SOC estimation in depth. In terms of SOH estimation of LIBs, Berecibar et al. (2016) reviewed the battery SOH monitoring methods, but it was not comprehensive. In Xiong et al. (2018), Xiong et al. (2018) divided the battery SOH estimation into experimental methods and model-based methods, while the benefits and drawbacks among the estimation methods were not discussed in-depth. To be specific, Li et al. (2019) presented the data-driven methods such as differential analysis and machine learning methods for battery SOH monitoring. However, a summary of the data-driven strategy was only provided, which is not enough for beginners (Li et al., 2019c). All in all, these reviews only summarize the progress of a single type of estimation, i.e., either SOC or SOH. In contrast, Hu et al. (Hu et al., 2019) (2019) and Wang et al. (Wang et al., 2020c) (2020) respectively presented an understandable review of state estimation technologies for BMS, in which both SOC and SOH estimation were summarized and discussed but not comprehensive. Due to the strongly correlated relationship between SOC and SOH, various joint estimation methods have been designed and proposed. However, the above reviews did not include the development in this research direction.

A variety of methods have been developed for both SOC and SOH estimation. Considering the practical applications, the methods can be roughly categorized into online and offline ones. The online methods can be used for the real-time state estimation of the battery. However, the offline methods are not suitable during battery operations due to strict experimental schemes or high computational costs. In this paper, we divide the estimation methods into two main groups as shown in Fig. 1, namely online estimation and offline estimation. In terms of online SOC estimation methods, they can be divided into three groups, including ampere-hour counting (AHC) method, model-based method, and data-driven method. In practice, the AHC method suffers from the initial SOC error and accumulative errors from measurement systems (Khan et al., 2021; Lin et al., 2021). Hence, it is not suitable for online applications for EVs due to the poor estimation accuracy. The open-circuit voltage (OCV) method is performed based on the functional OCV–SOC relationship. However, the OCV of the battery can only be measured after a long-time rest so that it is not suitable for the real-time SOC estimation. Moreover, the OCV exhibits differences between charging and discharging processes at the same level of SOC due to the hysteresis effects, which inevitably affects the SOC estimation accuracy (Xu et al., 2020a). In terms of SOH estimation, the online estimation methods are categorized into the differential analysis (DA) method, model-based method, and data-driven method. The offline methods consist of capacity measurement and internal resistance measurement in which capacity and internal resistance are the two main degradation parameters of the battery. These two degradation characteristics can be measured through specific tests to reflect the SOH status of the battery. For example, the capacity measurement needs to be discharged at a small discharge rate until reaching the cut-off voltage of the battery. As for the co-estimation of SOC and SOH, the current methods can be grouped into the model-based method, data-driven method, and advanced

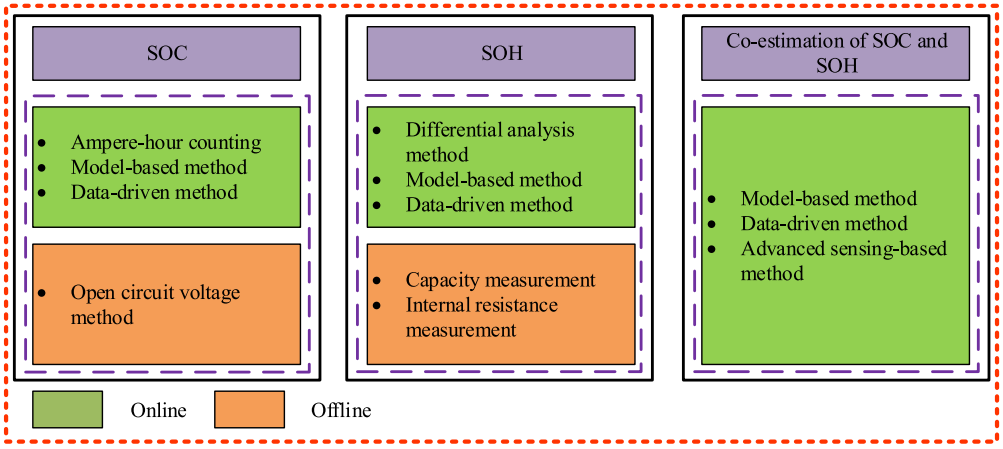


Fig. 1. Classification of SOC and SOH estimation methods.

sensing-based method. Specifically, the online methods for SOC and SOH estimation are discussed in this paper.

With the fast advancement in LIBs and EVs, more research work on advanced condition monitoring technologies are published, especially in recent three years. This makes the review work discussed above gradually not up-to-date. To bridge the research gap, this paper comprehensively discusses the state-of-the-art progress in SOC and SOH estimation of LIBs. Science Direct and IEEE are the main sources to search for relevant articles according to the keywords such as electric vehicles, lithium-ion battery, state of charge, and state of health. Compared with previous research work, the contributions of this review work are given as follows:

(1) The SOC/SOH estimation methods are divided into two categories, i.e. online and offline ones. The promising online estimation methods are specially discussed. The model-based method and data-driven method are mainly introduced for online SOC estimation. And the online SOH estimation includes (DA) methods, model-based methods, and data-driven methods.
(2) The existing online co-estimation strategies of both SOC and SOH are firstly discussed in this paper to fill the gaps in the research area of joint estimation. Then, it is reviewed from the aspects of the model-based methods, data-driven methods and advanced sensing-based methods.
(3) Based on the classification of state estimation, the latest research methods in recent years are selected and reviewed considering their strengths and drawbacks in practical applications.
(4) A list of key issues and future work are suggested for the advancement of online SOC and SOH estimation of LIBs.

The remainder of the rest is organized as follows: Section 2 briefly introduces the definition of the two important state evaluations. In Section 3, an overview of recent estimation technologies on SOC and SOH is given. Subsequently, Section 4 discusses the key issues and suggestions for further improvement. Finally, the overall conclusions of the research work are drawn in Section 5.

2. Definition of SOC and SOH

2.1. Definition of SOC

SOC is defined as the percentage of the remaining capacity to the maximum available capacity of the battery (Kim, 2008), and it can be given by

$$SOC\ (t) = \frac{C_r}{C_m} \times 100\% \tag{1}$$

where C_r stands for the remaining capacity that can be powered to electric devices. C_m specifically presents the maximum available capacity that the cell can store, which is determined by the electrochemical characteristics of the battery. The value of SOC ranges from 0% to 100%. A SOC of 0% denotes the battery is fully discharged, while a SOC of 100% means the battery is fully charged. In practice, the battery generally works under the SOC range from 20%–80% to avoid over-discharging (Wang et al., 2019a). In another way, SOC can be expressed by the Eq. (2) due to the relationship between the charging/discharging current and the battery capacity (Haisch et al., 2020).

$$SOC\ (t) = SOC\ (t_0) - \int_{t_0}^{t} \frac{I(t)\eta}{C_m} dt \tag{2}$$

where $SOC\ (t_0)$ and $SOC\ (t)$ represent the SOC at the initial time t_0 and time t, respectively. η denotes the coulombic efficiency that presents the ratio of the battery discharge capacity to the charge capacity during the same cycle. The current $I(t)$ varies with time in which it is negative in charging state and positive in discharging state. And a discrete form of Eq. (2) can be described as:

$$SOC_k = SOC_{k-1} - \frac{\eta \Delta T}{C_m} \cdot I_k \tag{3}$$

where ΔT is the sampling time, and I_k is the loading current. SOC_k and SOC_{k-1} represent the battery SOC at time step k and $k-1$, respectively.

In fact, the SOC values can be directly calculated when determining the initial SOC value according to Eq. (2) or Eq. (3). However, the inaccurate initial SOC value and the cumulative errors due to the measurement system can lead to significant estimation error in practical applications (Khan et al., 2021). Therefore, growing attention has been attracted to exploring the advanced methods for more reliable real-time SOC estimation.

2.2. Definition of SOH

Lithium-ion battery inevitably degrades with the increase of cycling and it consists of mechanical and chemical degradation. Mechanical degradation is mostly caused by the volume expansion or shrinkage due to the lithium de/-intercalation during the process of charging or discharging. Chemical degradation is mainly caused by electrolyte reduction and decomposition, solid electrolyte interphase (SEI) formation and so on since these processes can lead to the loss of lithium-ion and even the increase of the electrical resistance (Kabir and Demirocak, 2017; Xu et al., 2017). The degradation process of LIBs can be reflected by various

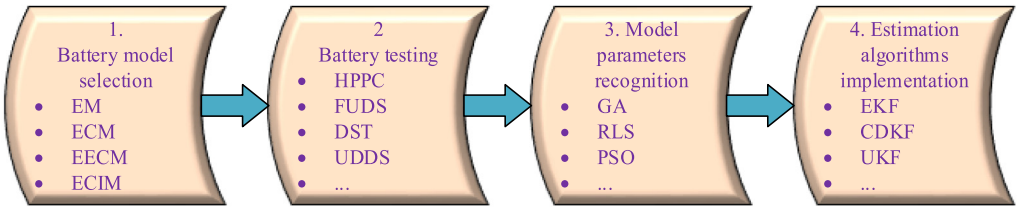


Fig. 2. The calculation process of model based SOC estimation.

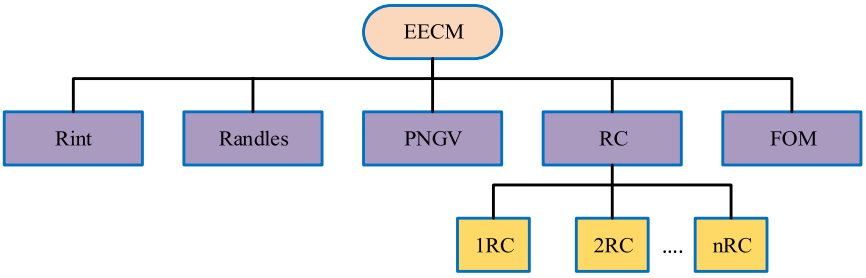


Fig. 3. Different EECM for modeling LIBs.

phenomena such as the attenuation of the maximum remaining capacity and the increase of the internal resistance. Two failure thresholds including an internal resistance increase of 100% and a capacity fade of 20% are regarded as the end of life (EOL) of the battery (Hu et al., 2019). This means batteries need to be replaced by new ones. Therefore, the SOH of the battery can be quantified by the ratio between the state value and the initial value of the capacity or internal resistance (Li et al., 2021b; Ge et al., 2021). They can be expressed as:

$$SOH = \frac{C_a}{C_{rated}} \times 100\% \tag{4}$$

$$SOH = \frac{R_{EOL} - R_{cur}}{R_{EOL} - R_{new}} \times 100\% \tag{5}$$

where C_a and C_{rated} are the actual and rated capacity, respectively. R_{cur} presents the current internal resistance through charging–discharging cycles. R_{EOL} and R_{new} are the ohmic internal resistance of a new battery and an EOL battery. Although the SOH monitoring needs to be tested and analyzed in a long-life period, developing accurate and effective SOH evaluation strategies is vital for replacement plans and fault detection of LIBs. Currently, neither capacity nor internal resistance is directly measurable with commercially available sensors, and they tend to be indicated and estimated through other measured variables such as the voltage, current and temperature. As a result, a variety of efforts are contributed to the effective SOH estimation of LIBs under different working conditions.

3. Overview of recent SOC and SOH estimation methods

3.1. SOC estimation methods

In recent years, a number of advanced techniques based on collected current and voltage parameters have been designed for real-time SOC estimation of LIBs. As discussed previously, the model-based and data-driven approaches for online SOC estimation are discussed in this section.

3.1.1. Model-based SOC estimation methods

As illustrated in Fig. 2, the model-based SOC estimation is generally carried out by four procedures: battery model selection, battery testing, model parameter recognition, and estimation algorithms implementation.

3.1.1.1. Battery model selection. The first stage is to select an appropriate model to simulate the electrochemical dynamics of LIBs. Four common models, including empirical model (EM) (Meng et al., 2018), electrochemical model (ECM) (Li et al., 2020e), electrical equivalent circuit model (EECM) (Mousavi and Nikdel, 2014), and electrochemical impedance model (ECIM) (Mu et al., 2017) were employed to replicate battery characteristics for SOC estimation. EM is used for SOC estimation by the empirical data fitting, which presents low accuracy to depict the dynamics of battery. Both ECM and ECIM require significant computational loads to solve the partial differential equations. The EECM is the simplest model that captures the highly dynamic behavior of LIBs through specific components such as voltage source, capacitors and resistors. By comparison, the EECM can achieve a better trade-off between estimation accuracy and computational efficiency. Therefore, the EECM for SOC estimation is specially discussed in this section.

Fig. 3 presents the major types of EECM: Rint model (Zheng et al., 2018a), Randles model (Gould et al., 2012), partnership for a new generation of vehicles (PNGV) model (Liu et al., 2016), nRC models (A review, 2021), and fractional-order model (FOM) (Lai et al., 2020c), and their circuit configurations are compared in Fig. 4.

- The Rint model combined with the OCV and ohmic resistance R_0 is regarded as the simplest EECM as shown in Fig. 4(a). It has lower computing complexity but with limited accuracy due to the rough simulation of LIB dynamics. Additionally, this model is unable to describe the critical hysteresis effect and polarization phenomenon that greatly influence the nonlinear performance of LIBs (Nejad et al., 2016). Hence, it cannot provide an overall study of LIBs characteristics, but it can be employed to explore local parameters like OCV due to its simplicity.
- Randles model shown in Fig. 4(b) is utilized to simulate LIB via treating the battery as a large capacity where C_2 is equivalent to OCV for storing charges. This model offers better simulating performance in lead–acid batteries (Calborean et al., 2019; Smith et al., 2019), and the simplified Randles circuit was used to establish equivalent impedance in LIBs (Nasser Eddine et al., 2018).
- In contrast, the PNGV and nRC models are developed to describe the nonlinear characteristics of LIBs. In case of

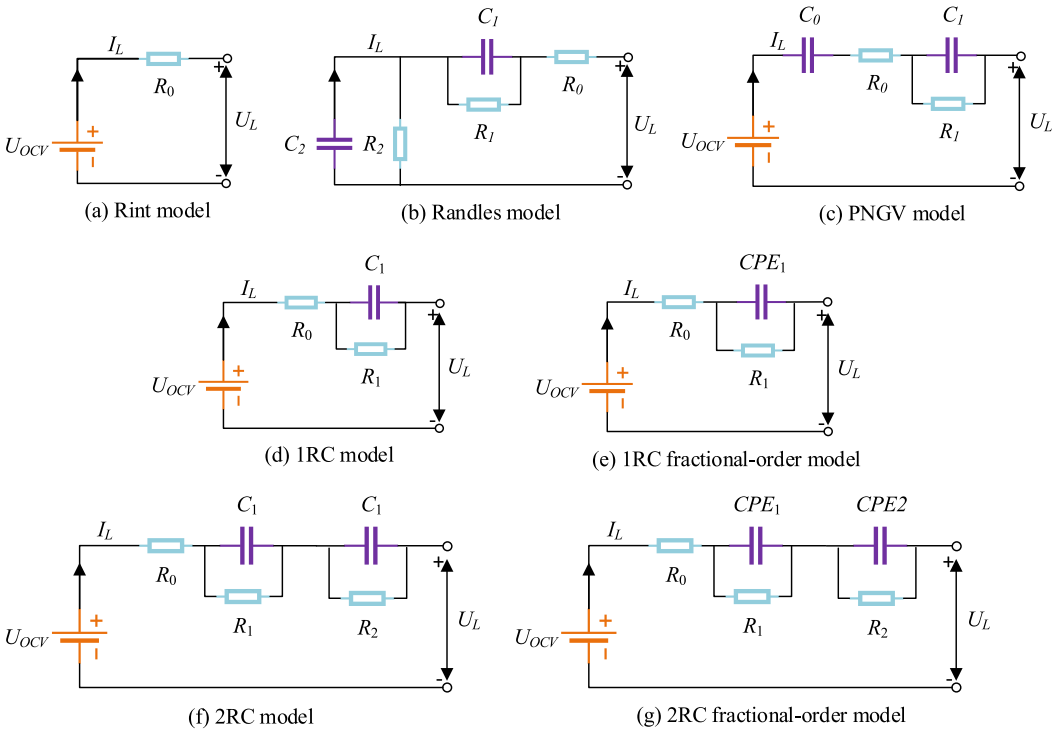


Fig. 4. Schematic diagram of different EECM models. (a) Rint model, (b) Randles model, (c) PNGV model, (d) 1RC model, (e) 1RC fractional-order model, (f) 2RC model, and (g) 2RC fractional-order model.

Table 1
Summary of popular EECM.

EECM	Ref. and Publication year
1RC	Wang et al. (2019a) (2019), Yu et al. (2017) (2017), Chun et al. (2018) (2018), Chen et al. (2019a) (2019), Peng et al. (2019) (2019), Liu et al. (2019a) (2019), Li et al. (2020a) (2020), Chen et al. (2021b) (2021), Jiang et al. (2021) (2021), Ben Sassi et al. (2020) (2020), Sandoval-Chileño et al. (2020) (2020), Hu et al. (2020) (2020), Shuzhi et al. (2021) (2021), and Loukil et al. (2021) (2021)
2RC	Guo et al. (2019) (2021), Xuan et al. (2020) (2021), Li et al. (2021c) (2021), Ouyang et al. (2020) (2020), Sun et al. (2020) (2020), Sarrafan et al. (2020) (2020), and Shrivastava et al. (2021) (2021)
2RCH	Lai et al. (2020b) (2020)
FOM	Lai et al. (2020c) (2020), Mawonou et al. (2019) (2019), Hu et al. (2018a) (2018), Lithium-ion (2019) (2019), Zhu et al. (2019) (2019), and Shen et al. (2018) (2017)

the PNGV model illustrated in Fig. 4(c), it incorporates an OCV source, an internal resistance, an RC branch and an additional capacity C_0 compared to the 1RC model as displayed in Fig. 4(d). The R_1 and C_1 are employed to present the polarization effect and the additional capacity C_2 can further improve the performance of the 1RC model (Shrivastava et al., 2019). Studies showed that PNGV presented better modeling performance in lower SOC areas (0%–20%) and the 1RC model had ideal identification in high SOC area (Lai et al., 2019). Moreover, nRC models consist of the combination of an internal resistance and a series of RC branches that can describe the transient response. It is noted that the increases of RC branches can better improve model accuracy, but the computational load increased as well (Mawonou et al., 2019).

- As can be seen in Fig. 4(e) and (g), FOM was respectively developed based on the 1RC and 2RC models and designed for accurately simulating the double-layer effect and solid-state diffusion of LIBs (Hu et al., 2018a; Xu et al., 2020b), in which CPE_1 and CPE_2 denote the constant phase elements.

Table 1 summarizes the popular applications of EECM in SOC estimation. It can be observed the 1RC model (Thevenin model) and 2RC model were widely used for SOC estimation of LIBs. By comparison, the 2RC model appeared to be the promising indicator for LIB SOC estimation due to the acceptable estimation accuracy and computation cost (Shen et al., 2018). In Zhao et al. (2017), it was found that the 2RC model could minimize the SOC estimation error to 2.3%, while the 1RC achieved an error as high as 6.2%. Moreover, Lai et al. (2020b) utilized the 2RC with one-state hysteresis (2RCH) model for SOC estimation and got better-simulated performance but with higher complexity. Furthermore, FOM is designed for SOC estimation and it has been validated that this model is able to reduce SOC estimation error and provide higher robustness compared to the classical 1RC model (Peng et al., 2019; Lithium-ion, 2019) and 2RC integral-order models (Hu et al., 2018a; Zhu et al., 2019). Overall, the EECMs such as the 1RC model and 2RC model gains popularity due to relatively high estimation accuracy and easy online application.

In addition, temperature has been found as an important influencing factor on the estimation accuracy of SOC. For example, the temperature can cause fluctuations in battery parameters such as OCV and internal resistance (Rui et al., 2011; Waag et al., 2013). Fig. 5 shows the OCV–SOC curves between 10%–100% SOC at different temperatures. It can be found that the OCV at the same level of SOC decreases gradually with the decrease of temperature, especially for the SOC between 10% and 70%. This phenomenon can be explained by the slow reaction

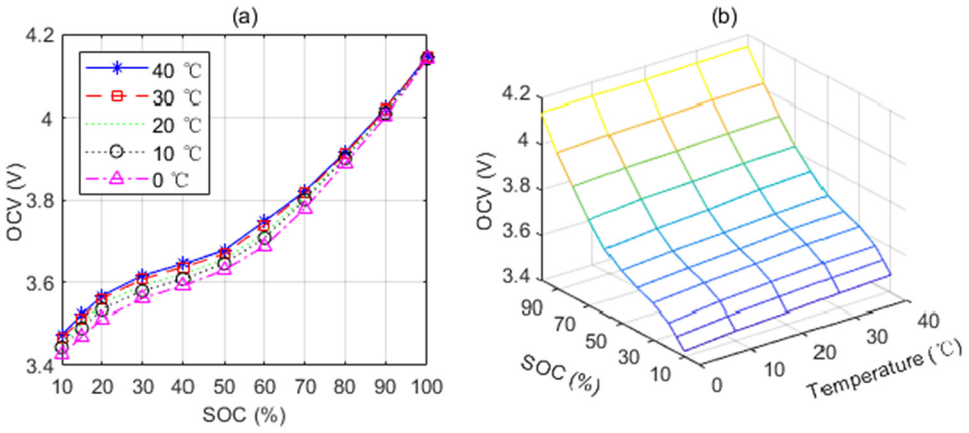


Fig. 5. (a) OCV–SOC profile obtained at different temperatures and (b) Curved surface of temperature based OCV–SOC..

of active materials inside the battery at lower temperatures. Therefore, the offline temperature-based OCV–SOC model is required to be developed for online model parameter updating and SOC estimation. For example, Liu et al. (2014) built the temperature-compensated model of OCV and internal resistance at the range from 0 °C to 50 °C with an interval of 10 °C based on the statistical data analysis to reduce the influence from battery temperature. Wang et al. (2015) established an OCV–SOC–Temperature mapping relationship from −20 °C to 60 °C to minimize the strong interference of temperature on SOC estimation. Xiong et al. (2020) established the temperature-dependent 1RC model covering -10 °C–40 °C based on a six-order polynomial function for model parameter identification and SOC estimation. The results show that the influence of temperature on OCV cannot be ignored. Compared with the fixed OCV–SOC relationship at different temperatures, the temperature-dependent model can significantly improve the SOC estimation accuracy. Feng et al. (2020) proposed an electrochemical–thermal–neural-network (ETNN) model in which the dynamic mechanism and temperature performance were integrated into the neural network to describe the electrochemical characteristics, thus reducing the impacts from the ambient temperature. In practice, the ambient temperature changes frequently, thus a non-linear relationship between temperature and OCV–SOC should be established for the accurate SOC estimation. Furthermore, the OCV–SOC curves can be affected by the battery SOH. The establishment of the OCV–SOC relationship should also consider the impact of battery SOH, while this is a time-consuming task.

3.1.1.2. Battery testing. After selecting an appropriate battery model, various types of tests can be carried out to obtain measurement data for model parameter recognition, such as hybrid pulse power characterization (HPPC) (Xu et al., 2020a), constant current discharge (CCD), federal urban driving schedule (FUDS), dynamic stress test (DST) (Hunt, 1996), urban dynamometer driving schedule (UDDS) (Duong et al., 2017), and Beijing driving cycle (BJDC) (He et al., 2013b).

- CCD is commonly used to estimate the variable capacity with terminal voltage or time, during which the effects of different current rates on the charging and discharging capacity of LIBs can be explored (Hunt, 1996).
- HPPC test contains a short pulse charge/discharge and resting stage, and it has been seen as an effective model parameter identification test since it is able to reflect the dynamics of the battery (Pan et al., 2020). Based on the voltage response curve obtained by the HPPC test, the observed hysteresis effect can be applied to recognize the polarization and ohmic resistance.
- FUDS denotes the variable power discharge scheme that accurately reflects the actual power conditions of EVs in practice.
- DST is developed into a simplified version of FUDS with the effective simulation of discharging process. Both the charging and discharging steps are included in DST and FUDS tests, while the main process is discharging phase.
- BJDC is another standard test scheduled to simulate the real operating environment of EVs.

Therefore, FUDS, DST, UDDS and BJDC tests conducted under complex working conditions can be utilized to assess the accuracy and robustness of the proposed methods used in the EV industry.

3.1.1.3. Model parameters recognition. Subsequently, the model parameters are recognized using specific identification methods based on the battery tests mentioned above. The model accuracy greatly relies on the precision of identified parameters, in which various parameter identification methods (PIMs) play an essential role in guaranteeing the estimation accuracy (Yang et al., 2018c). A variety of identification algorithms were put forward for accurate parameter recognition. For example, the widely used PIMs involve genetic algorithm (GA) (Zhang et al., 2017; Shen, 2018a), recursive least squares (RLS) (Snoussi et al., 2020; Boulmrharj et al., 2020), forgetting factor recursive least squares (FFRLS) (Yang et al., 2018a), particle swarm optimization (PSO) (Gao et al., 2018; Li et al., 2020b), and H-infinity filter (Shu et al., 2020c).

GA and PSO are two popular methods for solving the global optimization problems. It is noted that PSO was an ideal recognition method for the 2RC model and can also increase identification accuracy combined with the GA algorithm (Lai et al., 2019). However, these two methods can only be used offline for model parameter identification due to their iterative process. RLS method is the commonly used method for real-time parameter identification due to the fast calculation, while it needs to further confirm the least squares so that it is unable to effectively identify the model parameters. FFRLS (Yang et al., 2018a) was proposed to overcome this weakness and it improves the speed of convergence. Moreover, Zhu et al. (2020) presented RLS encounters the issue of identification biases due to measurement errors of current and voltage. Therefore, the recursive restricted total least squares (RRTLS) was proposed to reduce the identification biases, hence enhancing the identification accuracy of model parameters. In terms of model-based method, PIM has attracted increasing attention for accurate modeling since the accurate model parameter identification is the basis of the battery SOC estimation (Yang et al., 2020b).

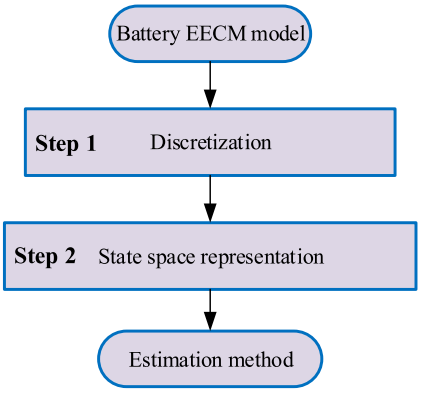


Fig. 6. The steps to establish a state space model.

3.1.1.4. Battery SOC estimation methods. The EECM cannot completely replicate the dynamic characteristics of the battery. Therefore, it is important to select suitable filtering algorithms to minimize errors between the actual data and the observed data for accurate SOC estimation, in which the computational complexity and estimation accuracy can be considered as the fundamental evaluation criterion. In order to achieve e online SOC estimation through combining with the selected EECM and filtering method, two crucial steps are required as shown in Fig. 6. The first step is to obtain the discretization model equations according to the determined battery EECM. In this step, the EECM is discretized according to the Kirchhoff voltage law and the SOC-OCV relationship. The second step is to establish state–space representation based on the model parameters identified by specific PIMs.

Here, an example of the 2RC battery model as shown in Fig. 4(f) is illustrated to the discretized model equations and state–space representation as expressed in (Shen et al., 2018).

$$U_1 = I_L R_1[1-\exp(-t/\tau_1)]$$
$$U_2 = I_L R_2[1-\exp(-t/\tau_2)] \tag{6}$$
$$U_L = U_{OCV} - U_1 - U_2 - I_L R_0$$

$$\begin{bmatrix} U_{1,k+1} \\ U_{2,k+1} \\ SOC_{k+1} \end{bmatrix} = \begin{bmatrix} e^{-t/\tau_1} & 0 & 0 \\ 0 & e^{-t/\tau_2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} U_{1,k} \\ U_{2,k} \\ SOC_k \end{bmatrix} + \begin{bmatrix} R_1(1-e^{-t/\tau_1}) \\ R_2(1-e^{-t/\tau_2}) \\ \dfrac{-\eta \Delta t}{3600 C_m} \end{bmatrix} I_k \tag{7}$$

$$U_{L,k+1} = U_{OCV}(SOC_{k+1}) - U_{1,k} - U_{2,k} - R_0 I_k \tag{8}$$

In the past few years, several methods have been proposed for SOC estimation. According to the statistical data from IEEE between 2009 and 2018 (Shrivastava et al., 2019), the KF family are the most popular online SOC estimation methods as shown in Fig. 7. Other major techniques including particle filter (PF) (Chin and Gao, 2018), slide mode observer (SMO) (Ning et al., 2018) and dual methods are also adopted for SOC estimation. KF-based methods highly depend on the battery model and system covariance (Shrivastava et al., 2019). The system noise is assumed as Gaussian distribution to solve the problem of error accumulation from the AHC method and improve estimation accuracy. However, the traditional KF method is a simple online estimation tool suitable for linear systems (Cheng et al., 2010), hence it cannot accurately reflect the nonlinear characteristics of LIBs.

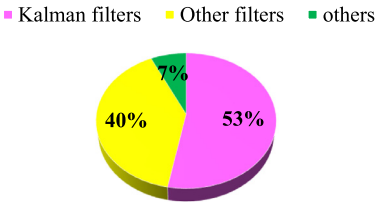


Fig. 7. The percentage of different SOC estimation methods between 2009 and 2018.

As summarized in Fig. 8, various nonlinear estimators have been proposed and widely used for online SOC estimation. The extended Kalman filter (EKF) (Lee and Kim, 2015; Zou et al., 2015) was proposed to improve the performance of linear KF by the Taylor-series expansion. However, EKF produces a large truncation error when linearizing the nonlinearities of the battery model (Waag et al., 2014). In order to overcome the deficiency in EKF, literature (Yang et al., 2018a) addressed the uncertainty of system noise using the adaptive extended Kalman filter (AEKF) and obtained satisfactory results in terms of SOC estimation. Based on the adaptive rule, AEKF can adjust the Kalman gains and covariance matrix, but it does not consider the change of error innovation sequence that changes because of system errors from the battery model and sensors. Therefore, Sun et al. (2021) proposed an intelligent adaptive extended Kalman filter (IAEKF) approach based on the maximum likelihood function to check and update the dynamic change of error innovation sequence for more accurate SOC estimation than AEKF.

As for the central difference Kalman filter (CDKF) (He et al., 2015), it was adopted to avoid the linearization error and improve the model precision for SOC estimation but with high computational load. Therefore, the CDKF with square root second-order difference transform (SRCDKF) was developed to avoid high order Taylor series expansion and complicated multi-parameter adjustment in other Kalman filters (Xuan et al., 2020). The experimental results revealed that the convergence of the SRCDKF algorithm was much quicker compared with traditional EKF.

Compared to EKF, the unscented Kalman filter (UKF) improves the accuracy of nonlinear fitting and reduce the complex computations based on the unscented transform. However, it is easily affected by the specified original value especially in a nonlinear system (Zhang et al., 2016). If a given initial state is far away from the real one, the prediction accuracy decreases, and even in some cases, the convergence will be lost. Therefore, some improved versions have been designed to address the drawbacks of UKF. An improved unscented Kalman filter (IUKF) was proposed in Chen et al. (2019c) to tackle the requirement for an accurate model and a priori noise statistics. Literature (Liu et al., 2019d) employed the adaptive unscented Kalman filter (AUKF) for SOC estimation in LIBs, which has the merit of adaptive correction of noise covariances in the process and measurement state. Experimental results show that the AUKF presents a good performance in convergence and estimation accuracy. However, this method needs to satisfy that the error covariance matrix is a positive definite matrix and non-positive definite error covariance matrix can lead to divergence, thus reducing the accuracy of SOC estimation. Therefore, Zhang et al. (2020a) proposed the improved AUKF based on singular value decomposition to address this issue and offer accurate online SOC estimation.

The cubature Kalman filter (CKF) method is applied to reduce the Gaussian noise through a third-degree spherical radial cubature rule, which is dedicated to dealing with dimensionality and divergence issues. It has been demonstrated that the CKF method shows better estimation and higher stability than EKF

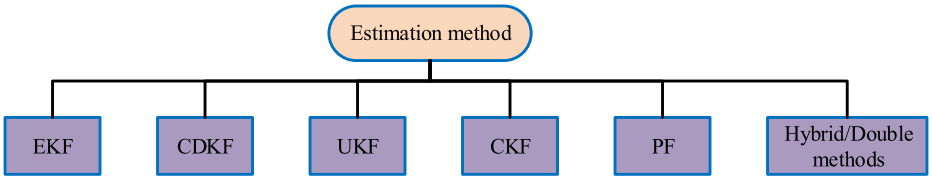


Fig. 8. Popular methods for online SOC estimation.

and UKF (Peng et al., 2019; Zeng et al., 2018). But it is prone to gross errors of observation noise and system noise (Cui et al., 2018). Compared to UKF and the adaptive cubature Kalman filter (ACKF) (Li et al., 2021a), the adaptive fifth-degree cubature Kalman filter (AFDCKF) solved the impact of large measurement error and initial error, hence receiving higher SOC estimation accuracy (Linghu et al., 2019). Meantime, the adaptive square root cubature Kalman filter (ASRCKF) (Yang et al., 2019b) improved the prediction precision by adaptively updating the noise covariance and high robustness against the measurement errors and parameter uncertainties. Therefore, it obtained accurate SOC estimation even if the method was initialized with inaccurate parameters.

To remove the limitation brought by the linear and Gaussian assumption, PF following specified distribution was employed for SOC estimation (Hao and Wu, 2015). The PF method aims to obtain a series of particles with relevant importance weights to present the posterior probability density (Shen, 2014). However, even after many calculations, traditional PF-based methods still cannot guarantee the global optimal value and its computational complexity is much higher than the classic EKF (Chen et al., 2019a). Hence, the unscented particle filter (UPF) method (He et al., 2013a) and improved adaptive particle filter (IAPF) (Ye et al., 2017) were proposed to suppress the measurement noise during the SOC estimation, while the computational complexity needs to be further reduced.

The estimators mentioned above have their own advantages and disadvantages, therefore a variety of hybrid applications have been developed to utilize their advantages and exhibited better overall performance. For example, one method can be used for model parameter identification and another for battery SOC state estimation. Yu et al. (2017) proposed the joint evaluation method in junction with H-infinity and UKF (HF–UKF) for online SOC estimation and received better estimation results under different operating temperatures. In order to improve the convergence ability and estimation robustness, Ye et al. (2018) employed the dual PF for the model parameter identification and reliable SOC estimation. Xu et al. (2020b) applied the dual Kalman filter (DKF) to SOC estimation, and the comparison results presented that SOC estimation error was within the range of $\pm 1\%$ under most test conditions. Besides, considering the influence of measurement noise on battery SOC estimation accuracy, the dual extended Kalman filter (DEKF) algorithm was proposed to reduce system noise and provided accurate SOC estimation (Lipu et al., 2020; Wang et al., 2019a). Liu et al. (2019a) designed a joint strategy combined with ACKF and singular value decomposition to reduce the observation error and nonlinear approximation error. Furthermore, Liu et al. (2019b) developed a hybrid cubature particle filter (CPF) that was able to offer more stable SOC estimation under harsh working conditions. It is noted that the system errors are unavoidable in the online SOC estimation and the errors may come from the model, voltage/current sensor, and estimation algorithm. In order to minimize the system errors, Lai et al. (2020a) proposed a hybrid SOC estimation method based on AHC and EKF, which specially considered the more reliable SOC increment to reduce the large errors from sensors and model.

Table 2 lists and compares the commonly used filtering methods for online SOC estimation. Compared to the application of the individual estimation method, the hybrid methods can give more robust battery SOC estimation. Therefore, more promising hybrid methods are worth exploring for SOC estimation in the future.

3.1.2. Data-driven SOC estimation methods

Compared with the model-based SOC estimation approaches, the data-driven methods are intelligent tools, free of considering the electrochemical dynamics of LIBs. Numerous data-driven strategies that focus on the relationship between input and output have been proposed for the SOC estimation due to the potential merits of high adaptability, nonlinear mapping, and flexibility (Dong et al., 2018). As shown in Fig. 9, the data-driven SOC estimation methods involve three major procedures: data collection, model training, and SOC estimation. They mainly focus on the discharging process based on the training features such as current, voltage and temperature.

Some popular intelligent methods, such as artificial neural network (ANN) (Ragone et al., 2021), support vector machine (SVM) (Meng et al., 2015), support vector regression (SVR) (Farmann et al., 2015), fuzzy logic (Ma et al., 2018), Gaussian process regression (GPR) (Deng et al., 2020) and GA methods (Chen et al., 2017), have been validated for SOC prediction. The traditional application of GA (Lu et al., 2018) cannot effectively evaluate the SOC, because it has slow convergence speed and cannot ensure converging to global optimization. Hence, the Chaos genetic algorithm (CGA) combining the global search ability was proposed to address the weakness of GA (El-Shorbagy et al., 2016). Moreover, an improved Chaos genetic algorithm (IGGA) was also designed for reducing the calculation amount (Shen, 2018b).

In addition, some advanced learning tools have recently been introduced for SOC estimation as shown in Table 3. Chemali et al. (2018) applied the deep feedforward neural networks (DNN) that self-learn their weights for SOC detection and offered better estimation performance. In Zhao et al. (2019), Zhao et al. introduced the combination of recursive neural network and convolutional neural network (RNN–CNN) with higher accuracy and faster convergence speed, in which RNN aimed to extract LIB status information that was seen as the input of CNN. Yang et al. (2019a) used the recurrent neural network with the gated recurrent unit (GRU–RNN) and received satisfactory estimation results under varying temperature conditions. Furthermore, Jiao et al. (2020c) proposed an improved version of GRU–RNN based on the momentum algorithm, which minimized the oscillation of the weight change in the gradient algorithm to improve the training speed and optimization process for reliable SOC estimation. However, it did not consider the influence of the ambient temperature. In order to eliminate measurement noise, Liu et al. (2019) feed some parameters obtained by KF to deep belief network (DBN) and improved the estimation accuracy. Similarly, Chen et al. (2019b) proposed the feedforward neural network (FFNN) that saved the historical information with EKF for accurate SOC prediction. In Xiao et al. (2021), a new deep learning method combined GPR with the gated recurrent unit kernel (GRU–GPR) was introduced for SOC estimation. The proposed deep learning kernel was applied to capture ordering matters and recurrent structures

Table 2
Summary of model-based methods for online SOC estimation.

Methods		Ref.	Key description
EKF	EKF	Chun et al. (2018), Chen et al. (2019a), Li et al. (2020a), Jiang et al. (2021) and Hidalgo-Reyes et al. (2020)	Suitable for nonlinear estimation but have high measurement error covariance
	AEKF	Yang et al. (2018a) and Lyu et al. (2019)	Process the uncertain states and reduce system error but ignore the distribution change of error innovation sequence
CDKF	CDKF	He et al. (2015)	High model precision but with high complexity
	SRCDKF	Xuan et al. (2020)	Suitable for non-negative covariance and low complexity
UKF	UKF	Li et al. (2017)	Reduce linearization error but is subject to priori noise statistics
	IUKF	Chen et al. (2019c)	Have adaptive noise distribution and fast convergence but high complexity
	AUKF	Liu et al. (2019d) and Du et al. (2014)	Adaption of the process and measurement noise but error covariance matrix must be positive definite matrix
CKF	CKF	Peng et al. (2019) and Zeng et al. (2018)	Low complexity but fixed noise covariance matrix is required
	AFDCKF	Linghu et al. (2019)	Reduce estimation error and fast convergence speed
	ASRCKF	Yang et al. (2019b)	High accuracy and reduce system error
PF	PF	Chen et al. (2019a), Hao and Wu (2015), Shen (2014) and Tulsyan et al. (2016)	Reduce system error but with high complexity
	UPF	He et al. (2013a)	Improve model precision but increase convergence speed
	IAPF	Ye et al. (2017)	Eliminate the estimation error but convergence speed is slow
Hybrid methods	HF–UKF	Yu et al. (2017)	Robust to inaccurate initial SOC value
	Dual PF	Lipu et al. (2020) and Ye et al. (2018)	Fast convergence speed
	Dual KF	Xu et al. (2020b), Guo et al. (2019) and Pavković et al. (2014)	High accuracy and fast convergence
	Dual EKF	Wang et al. (2019a) and Shuzhi et al. (2021)	Reduce system error but convergence speed is slow
	ACKF/SVD	Liu et al. (2019a)	Reduce system error
	Hybrid CPF	Liu et al. (2019b)	Reduce calculation error level and suppress particle degeneracy
	AHC/EKF	Lai et al. (2020a)	Suppress model and sensor errors

Table 3
Summary of data-driven methods for SOC estimation.

Method	Ref./Year	Input variables
DNN	Chemali et al. (2018)/2018	Voltage, current and temperature
RNN–CNN	Zhao et al. (2019)/2019	Voltage, current and temperature
GRU–RNN	Yang et al. (2019a)/2019	Voltage, current and temperature
Improved GRU–RNN	Jiao et al. (2020c)/2020	Voltage, current
DBN–KF	Liu et al. (2019)/2019	Voltage, current and temperature
FFNN–EKF	Chen et al. (2019b)/2019	Voltage, current and temperature
GRU–GPR	Xiao et al. (2021)/2021	Voltage, current and temperature
B-LSTM	Bian et al. (2020b)/2019	Voltage, current and temperature
Autoencoder–LSTM	Fasahat and Manthouri (2020)/2020	Voltage, current and temperature
LSTM and UKF	Yang et al. (2020a)/2020	Voltage, current and temperature
SBLSTM	Bian et al. (2020a)/2020	Voltage, current and temperature

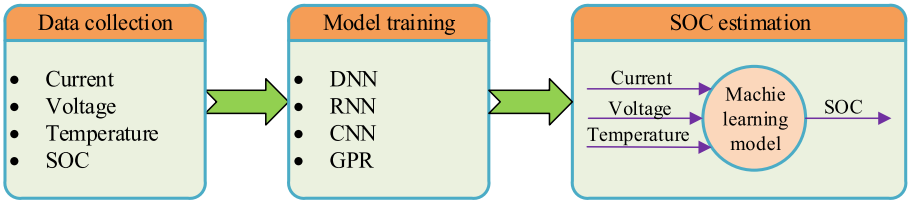


Fig. 9. The procedure of machine learning methods for SOC estimation.

in sequential measured data, which addresses the problem that the traditional kernel functions cannot consider the temporal dependence of training data. It is shown that the RNN methods present strong robustness against nonlinear dynamics, hysteresis, aging mechanism, and parameter uncertainties.

Long short-term memory (LSTM) neural network, as an improved RNN network in terms of gradient vanishing phenomenon, has recently received much attention in SOC estimation. In Bian et al. (2020b), the bidirectional long short-term memory (B-LSTM) was proposed for SOC estimation under different temperature conditions, which is capable to capture both historical and future measurement information and hence increases the estimation accuracy. Fasahat and Manthouri (2020) proposed the improved LSTM combined with an autoencoder (Autoencoder–LSTM) neural network in which the autoencoder neural network was used to extract and reconstruct the training features for LSTM. Experimental cases such as FUDS and DST revealed the effectiveness of this hybrid method compared to the multi-layer perceptron

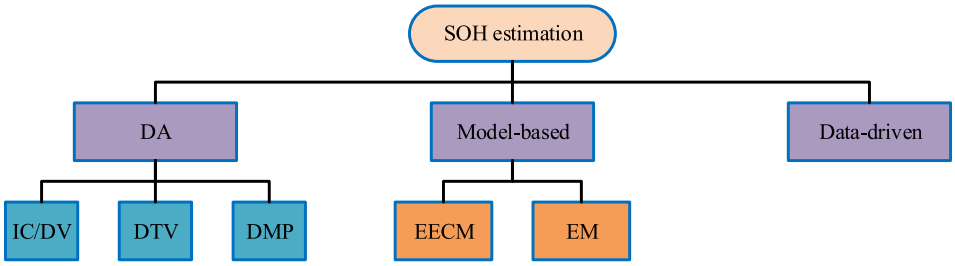


Fig. 10. Classification of online SOH estimation methods.

neural network. Moreover, Yang et al. (2020a) achieved accurate online SOC estimation based on LSTM and UKF. The LSTM model was offline trained using the measured current, voltage, and temperature from DST, FUDS, and US06 tests, and then the UKF was used to reduce estimation errors and realize online SOC estimation. Furthermore, Bian et al. (2020a) proposed a stacked bidirectional long short-term memory (SBLSTM) neural network for SOC estimation. The proposed method can capture the battery temporal contexts in both backward and forward directions and hence obtain the temporal dependencies from the past and the future information.

Although the current data-driven methods achieve the acceptable accuracy in SOC estimation, they mainly estimate battery SOC by capturing the nonlinear relationship between battery SOC and the electrical parameters such as voltage, current, and temperature as shown in Table 3. On the one hand, they do not consider the influence of the battery SOH, which causes inaccurate SOC estimation with the increasing aging or cycling. On the other hand, the training process of the data-driven method is time-consuming due to the input of a large number of training samples, which can possibly hinders their usage in practical applications.

3.2. SOH estimation methods

SOH estimation is another indispensable part of BMS, therefore, substantial efforts have been made to describe and predict SOH in LIBs. As shown in Fig. 10, these methods can be divided into three categories: differential analysis (DA) method, model-based method, and data-driven method.

3.2.1. Differential analysis methods

The DA methods are differential calculations based on voltage curves to get sensitive SOH-related features, and they can be categorized into four groups, including incremental capacity analysis (ICA), differential voltage analysis (DVA), differential thermal voltammetry (DTV) and differential mechanical parameter (DMP) estimation.

In the curve of ICA/DVA, it is found that the peak amplitudes decrease (Weng et al., 2013) and the observed peak positions change (Li et al., 2018) with the battery capacity fading. Based on the obvious peak features in the voltage regions, Li et al. (2019b) combined the gray relational analysis and the entropy weight method to further process the partial dQ/dV information for accurate SOH estimation. This method is only effective to analyze the battery SOH under low charge/discharge rates. To achieve more accurate evaluation of SOH, ICA/DVA tends to be excited with a small current rate. However, it may not be practical to ensure a low discharging current rate in some real-world applications (Xiong et al., 2018). Additionally, the peak is prone to be offset due to the obvious impedance change at high current rates (Shibagaki et al., 2018). These limitations have a large influence on accurate online SOH estimation. Therefore, it is important to reduce measurement noise by means of appropriate filtering and smoothing methods (Wang et al., 2017).

Moreover, DTV can offer complemental analysis in SOH estimation, which considers the temperature change with dT/dV and provides extra entropic features compared to ICA/DVA methods (Li et al., 2019c). Therefore, the dT/dV analysis with the extraction of voltage and temperature parameters has the ability to get more accurate analysis at high current rates than ICA/DVA (Merla et al., 2016a). In the case of dT/dV curve, the entropy information that presents the variation of peak height and positions is used to denote the increase of LIB impedance and degradation (Merla et al., 2016b). Although the DTV analysis can be achieved with low complexity, it is prone to environmental temperature, which can cause large diagnosis errors.

Furthermore, some mechanical parameters associated with cell SOH, such as stress and strain (ε) (Cannarella and Arnold, 2014), can be measured by load sensors mounted on battery surface to indicate SOH. In Sommer et al. (2015), it was demonstrated that battery SOH was linearly correlated with electrode expansion. A few studies on the first and derivative of strain to voltage ($d\varepsilon/dV$) (Sommer et al., 2015) and capacity ($d\varepsilon/dQ$) (Schiffer et al., 2015), and the second derivative of strain to capacity ($d\varepsilon^2/dQ^2$) (Oh et al., 2014) have been used for SOH estimation. The results are similar to that of ICA/DVA and show the phase transitions of both positive and negative electrodes. Based on the expansion identification, the DMP analysis can be applied to estimate SOH under low or high current rates. In practice, many batteries are constrained in a battery pack with limited space so that the cell expansion changes slightly, which makes it difficult to measure the actual cell swelling.

In summary, these DA approaches cannot guarantee precise online SOH estimation under different working conditions. As a result, they are likely to serve as complementary techniques for real-time SOH estimation.

3.2.2. Model-based methods

Model-based methods consist of three groups, i.e., EECM based methods, ECM based methods, and EM based methods. They are the commonly used online techniques for SOH estimation. The estimated aging-related features such as capacity degradation and resistance parameters can be used to reflect the battery SOH state.

The EECM is not only an important tool for SOC estimation, but also plays an essential role in SOH monitoring. Based on the established EECM, such as RC model (Andre et al., 2011) and FOM (Galeotti et al., 2015), some adaptively filtering algorithms such as, EKF (Tan et al., 2021), PF (Xiong et al., 2017), and AEKF (He et al., 2012), were employed to identify the electrical parameters such as resistance and capacity for the battery SOH estimation. Besides, the equivalent internal impedance was used to estimate SOH by considering the influences of temperature and SOC (Wang et al., 2019b).

Table 4
Summary of empirical models (EMs) for SOH estimation.

EM	Estimation factor	Discrete equations	Ref.	Estimation accuracy
Linear model	Capacity	$f(k,I) = \beta_1(k) + \beta_2(k) \cdot I + \varepsilon(k)$	Wang et al. (2020a)	R-square: 0.9778
Exponential model	Capacity	$lnf_1(n,t) = (\eta_1 + \eta_2 n)\exp(\eta_3 T) + \varepsilon_1(n)$	Wang et al. (2020a)	R-square: 0.9844
Hybrid model	OCV and capacity	$OCV(SOC,T) = \exp[\frac{E}{R}(\frac{1}{T_{ref}} - \frac{1}{T})] \cdot \sum_{k=0}^{m} a_{k,ref} SOC^k$	Bian et al. (2020c)	Relative error: less than 0.45%

In terms of ECM, it considers the electrochemical reactions which cause the degradation of lithium ions and the consumption of active materials, hence leading to the fading of battery life. Single particle model (SPM) is a classic electrochemical model in which the single particle is used to present the active material distribution inside the battery and hence the influence of solid-phase diffusion of the electrodes is investigated. (Wang et al., 2020c). However, SPM exhibits low accuracy to describe the electrochemical feature of the battery. On the basis of SPM, some improved models such as single particle model with electrolyte (SPMe) (Grandjean et al., 2019), enhanced single particle model (eSPM) (Sadabadi et al., 2021) and pseudo-two-dimensional model coupled with single particle model (P2D-SPM) (Bi et al., 2020) have designed for SOH prediction. In practice, the performance of these physics-based electrochemical models are limited in SOH estimation as lots of variables need to be identified based on optimization methods, which can easily lead to local optimization or over-fitting.

EM is another important model that is used to build the relationship between degradation factors and battery SOH. The specific empirical equitation is fitted through battery cycle tests in which the capacity fading is presented as the function of time or number of cycles. Table 4 shows a summary of fitting models which are established according to the cycle tests and have proven effective in SOH prediction. The aforementioned EMs especially consider the capacity loss and can receive faster calculations. In contrast, the introduction of the exponential model can achieve more accurate capacity fitting according to the estimation error. However, the current EMs still have many limitations in battery SOH estimation. For example, Wang et al. (2020a) respectively considered the relationship between discharge rate and temperature and capacity loss based on the linear model and exponential model, but the joint influence of these two factors on the battery degradation capacity is neglected. Bian et al. (2020c) proposed a hybrid model for SOH estimation, but the model is only suitable for the CC charge or discharge phase.

The EECM-based method considers the electrical characteristics of the battery, thus it can be easily carried out and extract the degradation feature. Specifically, the ECM-based method can reflect the detailed electrochemical process like the formation of the SEI and provide a more accurate SOH estimation. However, the complex electrochemical model probably causes a larger computation. Compared to EECMs and ECMs, EMs only consider the experimental data itself and do not consider the physical characteristics of the battery, which easily introduces larger estimation errors in practical applications.

3.2.3. Data-driven methods

Data-driven approaches based on machine learning techniques can give reliable SOH estimation since they do not need to simulate the complicated electrochemical model. In these methods, a set of aging related information are used as input to the intelligent model, and then the SOH or EOL is estimated and predicted. As presented in Fig. 11, there are four crucial steps in machine learning-based methods for SOH estimation. The first step is to collect data such as current, voltage, temperature and SOH. These simple parameters cannot guarantee the accurate identification of SOH, hence in step 2, it is highly necessary to further create and extract more high-quality features related to SOH based on the existing parameters. Subsequently, the machine learning model is trained in step 3 based on the selected features. Finally, SOH estimation is carried out in step 4.

Traditional machine learning algorithms directly regard the raw data such as voltage, current and temperature as inputs. However, the high nonlinear matching ability considerably depends on the aging data since the health indicators have a huge impact on the performance of SOH estimation. Therefore, extracting effective features correlated to health mechanisms and selecting appropriate machine learning methods are two crucial steps for effective SOH estimation in LIBs. Currently, various advanced methods combined with life-related feature extraction have been proposed and widely used for SOH monitoring as shown in Table 5.

First, the extreme learning machine (ELM) is a good choice to achieve SOH estimation. Pan et al. (2018) used ELM to develop the correlation between multiple health indicators and battery health conditions. This estimator presents higher estimation accuracy and efficiency than the traditional backpropagation neural network. Chen et al. (2021a) proposed a novel metabolic extreme learning machine (MELM) to present the complex battery degradation mechanism and achieve the SOH monitoring based on incremental analysis of degradation features for different types of batteries under different usage levels. The results show that effective SOH estimation is achieved thanks to the selection of the degradation features.

Second, SVR/SVM-based methods are good choices for SOH estimation of LIBs due to their fast computation speed. In general, the performance of SVR/SVM-based methods greatly depends on the setting of the initial parameters like the kernel parameter. Therefore, some optimization methods have been incorporated to determine the initial parameters of these types of methods for robust application. For example, the particle swarm optimization-least square support vector regression (PSO-LSSVR) approach (Yang et al., 2018b) was applied to offer a reliable SOH estimation in which the PSO was used to achieve global optimization of initial regularization parameter C and the kernel parameter γ. However, no obvious improvement has been obtained compared to the traditional LSSVR method. Similarly, GA was used to optimize the parameters of SVR in Cai et al. (2019) for SOH evaluation. However, the extraction of the degradation features is based on the pulse current discharging test, which cannot be applied to practical applications due to the complex working conditions in practice. In addition to discharging process, the constant current–constant voltage (CC–CV) charging profiles can also represent the degradation process and help the extraction of health indicators. In Deng et al. (2019), the least squares support vector machine (LSSVM) combined with a series of health features obtained from the charging curves was used for battery SOH prediction under different working conditions. In addition, Shu et al. (2020d,b) proposed a fixed size LSSVM

Table 5
Summary of data-driven methods for SOH estimation.

Methods		Ref.	Input features	Estimation error
ELM-based	ELM	Pan et al. (2018)	Terminal voltage, current, temperature, ohmic resistance and polarization resistance	RMSE: 1.09% MAE: 1.72%
	MELM	Chen et al. (2021a)	Mean ohmic internal resistance, mean polarized internal capacitance, and degradation capacity	MAE: less than 0.6% RMSE: 12.91%
SVR/SVM-based	PSO-LSSVR	Yang et al. (2018b)	Polarization resistance/capacity, ohmic resistance and SOC	RMSE: 3%
	SVR/GA	Cai et al. (2019)	Current/voltage from the key points in current pulse test	MSE: 3.6569×10^{-4}
	LSSVM	Deng et al. (2019)	Time interval of an equal charging current/voltage difference, charging capacity in constant current/voltage phase, temperature changing rate and average temperature.	RMSE: 3.42%
	Fixed size LSSVM	Shu et al. (2020d) and Shu et al. (2020b)	Charging voltage profile	MAX: 2%
	Ensemble learning/SVR	Meng et al. (2020)	Knee voltages of pulse response	MSE: less than 0.3%
ANN-based	ANN	Zhang et al. (2019)	Five features from different voltage regions in dQ/dV curves	RMSE: 5.41% MAE: 3.52%
	FFNN	Song et al. (2020b)	Charging capacity	MAX: 0.45%
	GRU–RNN	Fan et al. (2020)	Voltage, current, and temperature	MAX: 4.3%
LSTM-based	VLR-LSTM	Hong et al. (2021)	Ambient temperature and mileage	RMSE: less than 0.232% MSE: less than 5.38%
	AST-LSTM	Li et al. (2020f)	Voltage, current, and temperature and sampling time	RMSE: less than 2.7%
Transfer learning-based	Semi-supervised transfer learning	Li et al. (2020d)	Ratio of the CC mode, equal voltage drop, characteristics of IC curves, and sample entropy of discharge voltage	RMSE: less than 2.5% MSE: less than 0.5% MAE: 2%

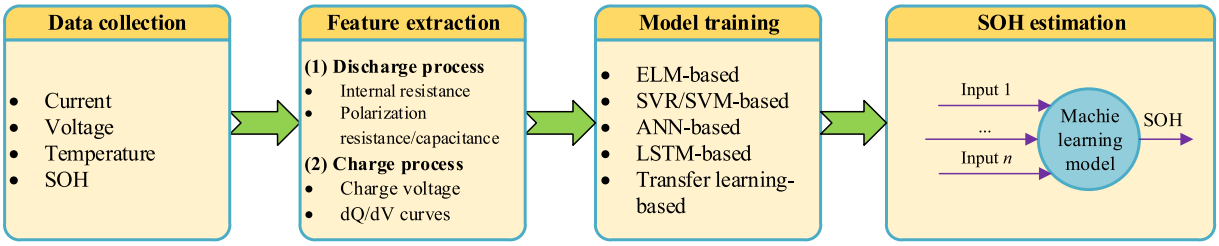


Fig. 11. The procedure of machine learning methods for SOH estimation.

method with mixed kernel function for online SOH estimation based on the charging voltage profiles. The charging experiments were carried out using typical CC–CV mode under fixed temperature. The experimental results highlighted the robustness of the proposed method. However, the training process does not consider the important influence of temperature, which cannot guarantee effective applications in practice. In order to ensure superior SOH estimation of LIBs in the energy storage systems, ensemble learning framework was employed to extract the high-quality health factors from a quantity of raw data and then SVR was used to learn and build the strong correlation between the extracted health indicators and battery life (Meng et al., 2020).

Moreover, a variety of ANN-based nonlinear estimators were widely used for SOH estimation of LIBs. Zhang et al. (2019) extracted features from the smoothed partial incremental capacity curves under CC discharging process and used the ANN model for SOH estimation. This training scheme is only suitable for the SOH estimation of EV charging systems working under CC discharging mode. Feedforward neural network was adopted to monitor battery SOH based on a huge quantity of actual EV battery data collected from the big data platform (Song et al., 2020b). This prediction model based on big data can be more effectively applied in practice than most of the models established based on static battery experiments. Based on the CC charging profiles of the Oxford Battery Degradation dataset (Birkl, 2017), the hybrid neural network-gate recurrent unit–convolutional neural network (GRU–CNN), was used for SOH estimation based on features including current, voltage and temperature (Fan et al., 2020). Fig. 12 shows the CC charging voltage data under different SOHs. It can be seen the terminal voltage exhibits differences at different life stages. As the battery degrades, the terminal voltage rises more quickly and the battery requires a shorter charging time to reach the cut-off voltage. Therefore, the health features can be extracted from the charging profiles. Compared to the discharge process, the charging process of the battery is more conducive to the selection of degradation features and the estimation of battery SOH. This is because the battery frequently works under complex discharging conditions, while it is charged under CC mode.

LSTM, as a variant of RNN, has also attracted extensive attention for SOH estimation. To address the complex driving behaviors and operating conditions during real-world environments,

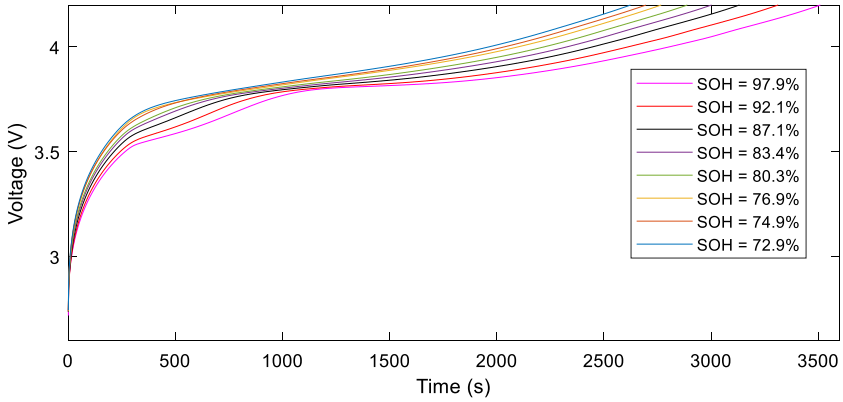


Fig. 12. Charging voltage profiles with time under different SOHs.

Hong et al. (2021) proposed the variable-length-input long short-term memory (VLR-LSTM) network to learn the relationship between battery SOH and degradation features extracted from the big data platform. Results show this method can be applied to full-climate and full-state EV applications. Also, a variant of LSTM (AST-LSTM) was designed in Li et al. (2020f) to improve SOH estimation based on the NASA dataset. In this method, the constant error carousel was added to the output gate to remove the unwanted error information so that the more useful contents from old and new signals can be extracted for SOH prediction.

These machine learning methods require significant amount of aging-related data to guarantee reliable model training and accurate SOH estimation, which inevitably requires a large amount of time and economic cost. In order to address this issue, Li et al. (2020d) developed a semi-supervised transfer learning method to estimate the SOH degradation trends for batteries under different working conditions. The root-mean-squared error (RMSE), mean absolute error (MAE), maximum relative error (MAX) and mean square error (MSE) are critical metrics used to evaluate the error between real SOH and estimated SOH. It is concluded that it is quite important to select high-quality input characteristics related to the aging mechanism for precise SOH estimation. Meanwhile, the parameter extraction needs to consider the complexity of the online estimation operation.

3.3. Co-estimation methods of SOC and SOH

Apart from the pure SOC or SOH estimation, it is technically challenging for the simultaneous estimation of these two key state variables. The variation of some important factors can indicate SOC in a short timescale and reflect SOH in a long timescale. Based on the close correlations between SOC and SOH, the joint estimation is likely to be achieved. The co-estimation strategy has the advantage to improve prediction accuracy because it focuses on the mutual effects from different battery states. A lot of research works have been devoted to reliable co-estimation of SOC and SOH.

ECM provides a choice for joint SOC and SOH estimation in which the lithium-ion concentration phenomenon is considered. Liu et al. (2020) developed the joint estimation based on the simplified P2D model by considering the electrolyte characteristics and dependency of surface and bulk lithium-ion concentration in the solid phase. And PF method was used to estimate the average lithium-ion concentration for SOC monitoring and the battery SOH can be calibrated according to the predicted average lithium-ion concentration at cut-off voltages of both charging and discharging stages. This method offers a good perspective on joint estimation and the maximum estimation errors of SOC and SOH were limited to 2% and 2.8%, respectively.

Moreover, various methods based on EECM have been employed for the co-estimation of both battery SOC and SOH. Because the model-based method is an effective path for SOC estimation and the identified model parameters present a high relationship with battery life span. For example, Topan et al. (2016) carried out the joint estimation combined with the 1RC model and KF algorithm, while the mean relative error was as much as 5.26% as the linear estimator KF easily introduces large system errors. In order to avoid the uncertain factors such as modeling, parameter error, and measurement noise, Afshari et al. (2018) used the smooth variable structure filter (SVSF) and 3rd-order RC model for SOC and SOH estimation. The results reveal that the SVSF achieve more accurate SOC estimation than EKF, however the battery SOH can only be roughly estimated based on the SVSF's chattering indicator. Wassiliadis et al. (2018) utilized the 2RC model for co-estimation of SOC and SOH combined with the DEKF method, while the internal resistance for SOH monitoring was easily influenced by the environmental temperature that had a huge effect on accurate SOH estimation. Considering the influence of temperature on joint estimation (Shu et al., 2020a), the relationship among OCV, SOC, SOH and temperature was embedded in a 2RC model. In this framework, AEKF combined with FFRLS was firstly used for SOC estimation and the identified model parameters such as internal resistance, polarization resistance, OCV, etc., were employed for SOH estimation. To improve uncertainties in the joint estimation process, Song et al. (2020c) introduced the novel sequential algorithm for crucial parameter identification. In this method, the current with different frequencies were injected, and then the ohmic resistance and RC parameters in 1RC model were determined accurately. Experimental results presented better estimation performance when the model parameters were evaluated for simultaneous SOC and SOH estimation. In addition, Qiu et al. (2020) proposed a hybrid method including backward smoothing square root cubature Kalman filter (BS-SRCKF) and EKF for joint estimation of SOC and SOH based on the 2RC model. Then the estimation performance was improved using the optimized cuckoo search algorithm in combination with the PF method. The results showed the proposed hybrid method gave higher effectiveness compared to traditional PF and UPF methods. However, the mixed application of too many methods can lead to a huge computational burden. In Li et al. (2019a), the RLS and AEKF methods were used for real-time model parameter identification and SOC estimation based on the 1RC model, and the battery SOH was identified via Elman neural network combined with partial charging voltage curves. However, this combination of model-based and data-driven estimation methods can undoubtedly increase the computational power in practical applications. The traditional onboard BMS has limited storage capacity and processing speed.

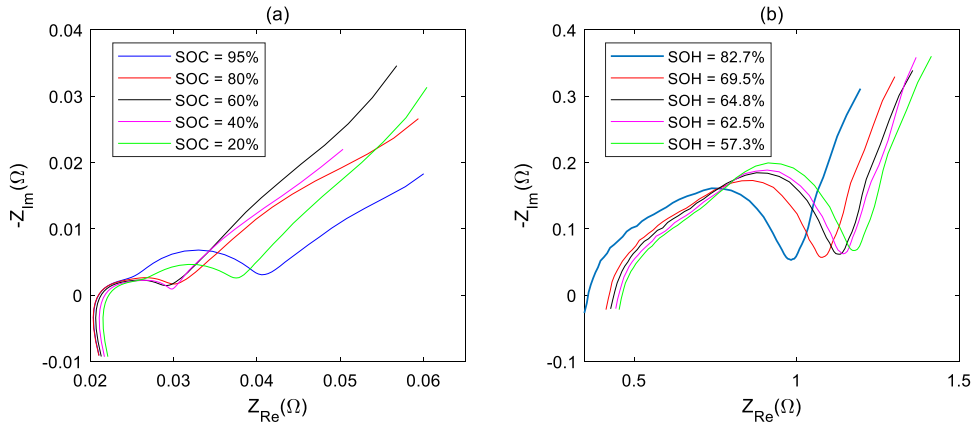


Fig. 13. EIS spectrum at different (a) SOCs and (b) SOHs.

In Li et al. (2020c), a cloud-based joint online estimation strategy was explored based on 2RC model to overcome the increasing computation and data storage pressure of onboard BMS. In the method, an adaptive extended H-infinity filter was used for reliable SOC estimation and the PSO-based method was applied to SOH monitoring based on the close relation between battery capacity degradation and internal ohmic resistance increment.

Model-based methods inevitably require complex testing conditions and procedures for joint estimation where it is an important part to guarantee the accuracy of the identified model parameters or indicators, otherwise, it can strongly influence the estimation performance of both SOC and SOH. Free of considering the dynamics and nonlinear electrochemical characteristics, the data-driven model can be used for the battery SOC and SOH estimation. In Song et al. (2020a), LSSVM was proposed to build the mapping model for SOC and SOH estimation based on the measured current and voltage under different time scales. And the UPF algorithm was implemented to optimize the results obtained by LSSVM estimation to further minimize the estimation errors.

Furthermore, some methods based on advanced sensing technology, such as ultrasound inspection and electrochemical impedance spectroscopy (EIS), can put insight into the internal structure of the battery. They have been successfully applied to the joint estimation of the battery. On the one hand, it is concluded that the properties (density and elastic modulus) of battery active materials change during cycling and aging (Davies et al., 2017), which can affect the propagation of the ultrasonic wave through the medium of the battery. The acoustic impedance can be considered as the function of variables SOC and SOH. For instance, the acoustic impedance increases with the discharge of the battery while decreases with the battery degradation (Ladpli et al., 2018). Therefore, ultrasonic analysis has been used for the monitoring of SOC and SOH. It was found the variation in time of flight and signal amplitude in the ultrasonic signal correlated heavily with the SOC and SOH of the battery (Davies et al., 2017; Ladpli et al., 2018; Copley et al., 2021). On the other hand, EIS is a non-destructive test method that can obtain the impedance of the battery using external signal excitation (Meddings et al., 2020). With the cycling and the aging of the battery, the material properties change and cause the variation of the battery impedance. According to the EIS dataset in Kollmeyer (2018) and Zhang et al. (2020b), Fig. 13 displays the EIS spectrum of the battery at different SOCs and SOHs. It can be found the spectrum presents differences at different SOCs and SOHs. Therefore, EIS measurement can be used to simultaneously estimate the battery SOC and SOH. The authors in Babaeiyazdi et al. (2021) and Kim et al. (2020) presented the cell impedances are correlated with battery SOC and SOH.

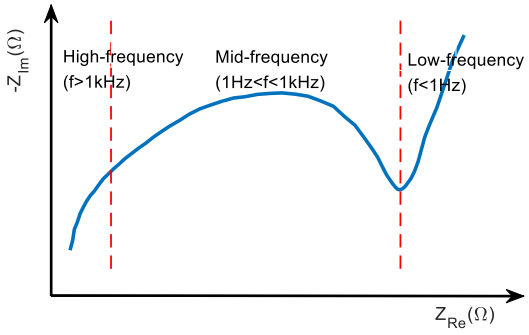


Fig. 14. Typical EIS spectrum of the battery obtained over a range of excitation frequencies.

In contrast, the model-based and data-data-driven methods for both SOC and SOH estimation highly depend on the measurement of the electrical parameters, such as current, voltage, and temperature. This can lead to restricted accuracy in estimation of the battery states (Popp et al., 2020). It has been validated that the ultrasound-based method and EIS measurement-based method can put insight into the internal structure of the battery so that they can achieve a more rapid state estimation of the battery. However, the requirements for the specialized measurement system may prevent them from being embedded in the BMS, which makes application at the battery pack level more difficult. Experimental results are susceptible to experimental conditions, which leads to poor experimental repeatability. In terms of the ultrasound-based method, the selection of transducer with specific resonance frequency should be investigated to obtain a stronger correlation between battery states and characteristic parameters. Fig. 14 illustrates the EIS spectrum over a range of excitation frequencies. It is time-consuming to obtain the full spectrum by the excitation under a large number of frequencies. Therefore, it is highly necessary to conduct the sensitivity analysis to select the effective excitation frequency for the reliable battery state estimation.

4. Key issues and future work

According to the comprehensive review in Section 3, a variety of techniques have been proven effective in estimating the SOC or SOH state of LIBs. However, substantial progress is still required to improve estimation accuracy and computation efficiency for online applications. As a dynamic coupling system, many factors restrict the effective monitoring of SOC and SOH in practice.

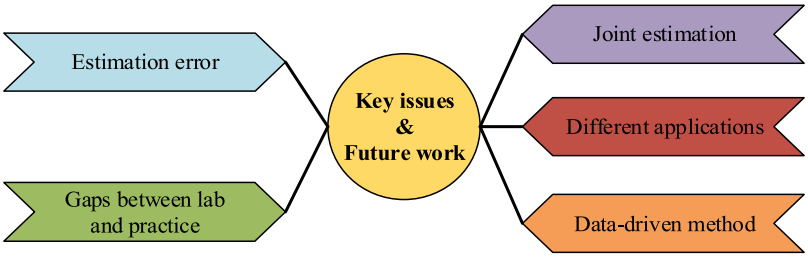


Fig. 15. Key issues and future work for online SOC and SOH estimation.

Based on the investigations, the research issues and future directions in online SOC and SOH estimation are outlined from five perspectives as illustrated in Fig. 15.

4.1. Estimation errors

Several error sources can influence the state estimation process of LIBs, which can come from the battery model, measurement system, estimation algorithms and so forth. These errors are not negligible as the accumulation of these errors will eventually cause significant estimation errors. Firstly, it is impossible for any model to entirely replicate the nonlinear behavior of LIBs. For example, the hysteresis effect increases the uncertainties of modeling. Therefore, a more accurate model needs to be built based on the genetic multi-physical modeling method combined with thermal, electrochemical, and series–parallel circuit models to replicate the dynamics of the battery. Secondly, in the case of model parameter identification, inaccurate parameters can impair the state estimation performance. Although the existing PIMs such as PSO and RLS have proven effective, the more precise model parameters can be obtained through using the optimized versions which consider the physical characteristics of the battery. Thirdly, there is measurement noise from sensors such as current, voltage, and temperature. Although these errors tend to be small, they should be minimized as the error accumulation can cause the drift of battery state estimation. To some extent, measurement errors can be eliminated by improving experimental conditions. Finally, the estimation methods also introduce the process and measurement noise during the online applications as discussed previously. Thus, the existing estimation methods such as AHC and filtering technologies need to be improved or combined to minimize the system errors for accurate SOC or SOH estimation.

4.2. Gaps between lab and practice

Currently, most of the research work for the battery SOC and SOH estimation is still in the laboratory stage. Some influencing factors such as changing ambient temperature and computational efficiency greatly influence the battery SOC and SOH estimation in practice. The LIBs in EV or charging systems work under complex operation conditions where the ambient temperature varies frequently. Due to the electrochemical dynamics of LIBs are easily affected by the temperature, it produces a significant gap between laboratory research and practical application. Therefore, the changing thermal information should be considered in battery modeling or state estimation methods, especially for model-based methods. For model-based methods, the model parameters must be updated in time under dynamic operating conditions. Subsequently, the real-time parameter update will undoubtedly increase the computational complexity of BMS. Therefore, it is necessary to conduct the sensitivity analysis of battery model parameters under different SOC or SOH to determine the crucial parameters that are highly sensitive to state estimation accuracy. Other insensitive parameters can be kept constant or updated less frequently, which will reduce the computing load of the online application. As a result, future work should focus on the SOC and SOH estimation considering the important factors of temperature and computational load, which facilitates the engineering application of laboratory methods in practice.

4.3. Joint estimation

A variety of advanced techniques have been proposed for individual SOC or SOH estimation. However, limited works have focused on the effective joint estimation of SOC and SOH. The relatively accurate results can be only achieved if estimating one of them separately and not considering another one. Because the battery is a dynamic system in which multiple states are coupled with each other like the relationship between SOC and SOH. Specifically, accurate SOC prediction should be associated with the variation of SOH as the capacity degradation also influences the parameters in the model-based methods for SOC estimation. In addition, the reliable SOH estimation can provide an accurate initial value for SOC monitoring. Note that the joint estimation can increase more computational load compared to one state estimation. The accurate co-estimation of both SOC and SOH is a promising but challenging work. As a result, the more accurate and computationally efficient applications of advanced methods become the future direction.

4.4. Different applications

Currently, the LIBs are widely applied to EVs and EV charging systems owning to their high energy density and reliability. Although a number of methods are specifically developed for the state estimation of LIBs, most of the advanced techniques exhibit poor generality if they are applied to different use of LIBs. On the one hand, the existing methods pay more attention to the cell battery instead of the battery packs or battery module. Addressing the inconsistency of the batteries and providing accurate battery pack or module state estimation have realistic meanings in practice. On the other hand, the LIBs in EVs and EV charging systems hold different working conditions, hence the effective estimation methods or PIMs should be explored to cover these different applications. Furthermore, when LIBs start to reach their EOL due to capacity and power degradation, those retired batteries be assembled for secondary applications, such as light EVs and distributed energy storage systems. The degradation of the battery and the second assembly of a series of retired batteries increase the uncertainty and instability of the battery operation. Hence, it is essential to develop reliable condition estimation methods for the secondary use of retired batteries in different fields. In short, substantial research efforts are still required to improve the reliability and accuracy of state estimation methods on different applications of LIBs.

4.5. Data-driven method

With the advancement of cloud computing technology and the availability of large amounts of monitoring data, data-driven techniques have received increasing attention for the state estimation of LIBs. Compared to the model-based method, the data-driven approaches can better map the nonlinearity based on self-learning characteristics. To achieve good performance in practical applications, two critical directions including feature selection and algorithm improvement must be enhanced to improve the robustness and reliability. On the one hand, intelligent cloud computing technologies based on the real-world battery data collected from the big data platform can be developed to achieve real-time and effective monitoring of a large number of batteries in practice. Cloud-based machine learning approaches can solve the problem that limited data is derived from a single vehicle or battery module. In addition, the real-world data covers complex operating conditions, which can facilitate more effective state prediction in practice. On the other hand, the performance of machine learning methods is heavily sensitive to the quality of training features. Therefore, it is very important to extract effective features from various sensing signals such as current, voltage, temperature, acoustic-ultrasonic, and EIS under charging or discharging phases. To achieve online estimation, the computational load for SOC and SOH estimation needs to be reduced. This requires the number of selected training features should be small and hence it is highly necessary to develop advanced online intelligent learning algorithms with the requirement of small-size data. It might be a good practice to combine the intelligent methods with the battery model to cover the battery dynamics, thus improving the estimation accuracy. As a result, data-driven methods should be further developed to promote the application of machine learning techniques in practice.

5. Conclusions

In this paper, the promising online SOC and SOH estimation technologies for EV batteries are reviewed with a focus on the development in recent five years. Model-based and data-driven methods are specifically reviewed for online SOC estimation. The DA, model-based, and data-driven methods for real-time SOH estimation are investigated, respectively. In addition, the co-estimation of SOC and SOH based on model-based, data-driven, and advanced sensing-based methods are reviewed. These methods are discussed and evaluated by focusing on their strengths and drawbacks. Finally, the key issues and future work are proposed for the guidance of online SOC and SOH estimation of LIBs in practice.

The current online estimation methods still face significant challenges when applied to practice. On the one hand, large errors from the battery model and measurement systems have significant influences on the state estimation of the battery. Moreover, these methods are generally developed based on laboratory data, thus causing a large gap between lab and practical applications. The battery is a time-varying and nonlinear electrochemical system so that its status can be easily affected by various factors such as ambient temperature and charge–discharge rate, which increases the difficulty of state estimation in practice. Furthermore, model-based and data-driven methods suffer from complex computations, which undoubtedly increases the computation load, especially when applied to battery packs.

To overcome the challenges, this review also gives the suggestions as follows:

(1) More accurate model can be built based on genetic multi-physical modeling method combined with equivalent circuit model, temperature and electrochemical to adapt to the complex external environment in practice.
(2) Nonlinear estimators and parameter identification methods need to be improved in terms of practical applications, estimation accuracy and computational efficiency.
(3) The joint SOC and SOH estimation methods can be developed to improve the estimation accuracy. Specifically, the ultrasound-based and EIS measurement-based methods should be further explored to achieve rapid state estimation.
(4) The estimation methods should be developed to facilitate the batteries in different applications such as the cell battery or battery pack in EVs and EV charging systems.
(5) In terms of data-driven methods, the effective feature selection and prediction methods based on small sample data should be explored to improve the estimation accuracy and efficiency. And the data-driven methods based on the big data platform are desired to be developed to facilitate practical applications.

In summary, the development of the SOC/SOH estimation considering the practical applications of LIBs is still a hot research topic. Different estimation methods can be specifically selected for various application scenarios. The proposed key issues and suggestions can hopefully benefit future research, development, and application of practical BMS.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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